Assignment 1

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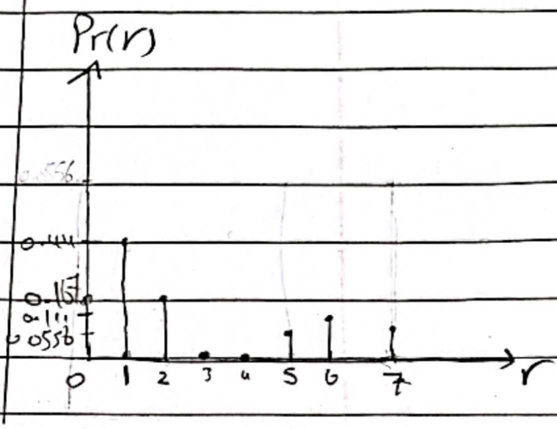
Sec.2

Question 2 :

r_k	n_k	$P_r(r_k) = n_k/MN$ (18)
$r_0=0$	3	$\frac{3}{18} = 0.167$
$r_1=1$	8	$\frac{8}{18} = 0.44$
$r_2=2$	3	$\frac{3}{18} = 0.167$
$r_3=3$	0	0
$r_4=4$	0	0
$r_5=5$	1	0.0556
$r_6=6$	2	0.111
$r_7=7$	1	0.0556



$$\boxed{s_k = T(r_k) = \frac{L-1}{MN}\sum_{r=0}^{k} n_i}$$

$s_0 = 7 \times 0.167 = 1.169 \simeq 1$

$s_1 = 7 \times (0.167 + 0.44) = 4.249 \simeq 4$

$s_2 = 7 \times (0.167 + 0.44 + 0.167) = 5.418 \simeq 5$

$s_3 = 7 \times (0.167 + 0.44 + 0.167 + 0) = 5.418 \simeq 5$

$s_4 = 7 \times (0.167 + 0.44 + 0.167 + 0 + 0) = 5$

$s_5 = 5.8072 \simeq 6$

$s_6 = 6.5842 \simeq 7$

$s_7 = 6.9734 \simeq 7$

4	3	4	4	5	1
1	4	6	4	1	4
4	7	7	7	9	5

$0 \Rightarrow 1$

$1 \Rightarrow 4$

$2 \Rightarrow 5$

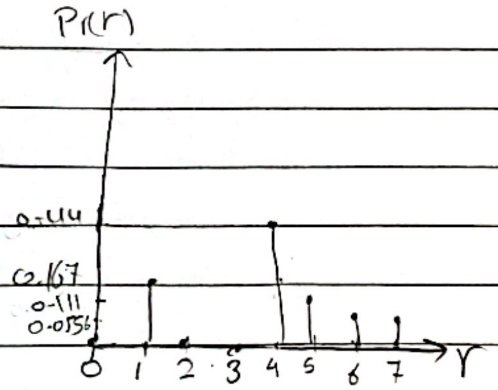
$3 \Rightarrow 5$

$4 \Rightarrow 5$

$5 \Rightarrow 6$

$6 \Rightarrow 7$

$7 \Rightarrow 7$



Sheet 1

Question 3

a) Discuss the different computation levels.

① Point level : $F_B[i,j] = O_{point}\{F_A[i,j]\}$

this means that the intensity of each pixel in the output image only depends on the intensity of the corresponding pixel in the input image.

② Local level : $F_B[i,j] = O_{local}\{F_A[k,l] ; [k,l] \in N[i,j]\}$

where $N[i,j]$ is a neighborhood around the position $[i,j]$, the intensity of each pixel in the output image depends on the intensity of pixels in the neighborhood of the corresponding position in the input image.

③ Global level : $F_B[i,j] = O_{global}\{F_A[k,l] ; 0 \le k < m , 0 \le l < n\}$

for an m×n image

the intensity of each pixel in the output image may depend on the intensity of any pixel in the input image

④ object level :

operations must be performed at the object level, that is include all pixels that belong to a particular object

b) Discuss the image types:

① Binary Images:
A binary image contains only 2 possible pixel values: 0 (black) and 1 (white).

② Gray scale Images:
- A grayscale image represents intensity only, pixels values usually range from 0 to 255 where 0 = Black and 255 = White.

③ Color Images (RGB images)
- A color image is composed of three channels: Red, green, Blue, each channel has values from 0 to 255, each pixel uses 24 bits (8 bits per channel

c) Define the following keywords:

I) Computer vision:
is the science of building systems that can extract certain task-relevant information from a visual scene.

II) Pixel:
- A pixel is the smallest addressable element of an image that stores intensity or color information at a specific position

III) Gamma correction:
variety of devices used for image capture, printing and display respond according to a power law. the process used to correct this power-law response phenomena is called gamma correction.

IV) Piecewise-Linear Transformation functions and it's usage.
is an intensity mapping technique that uses multiple linear segments to modify pixel values mainly for contrast stretching, Gray-level slicing and Bit plane slicing.

VI) Histogram equalization:

Histogram equalization is a technique that enhances image contrast by transforming pixel intensities to achieve a more uniform intensity distribution.

VII) Contrast stretchi

one of the simplest piecewise linear functions is a contrast-stretching transformation, which is used to enhance the low contrast images.

Question 4:

Gray level	0	1	2	3	4	5	6	7
No. of pixels	2	3	2	1	2	3	2	1

① Total no. of pixels:

$$N = 2+3+2+1+2+3+2+1 = 16$$

② propability of each gray level: $p(g) = \frac{n(g)}{16}$

$p(0) = \frac{2}{16} = 0.125$, $p(1) = \frac{3}{16} = 0.1875$, $p(2) = \frac{2}{16} = 0.125$

$p(3) = \frac{1}{16} = 0.0625$, $p(4) = \frac{2}{16} = 0.125$, $p(5) = \frac{3}{16} = 0.1875$

$p(6) = \frac{2}{16} = 0.125$, $p(7) = \frac{1}{16} = 0.0625$

③ Total mean gray level

$$\mu_T = \sum g.P(g)$$

$$\mu_T = 0\times0.125 + 1\times0.1875 + 2\times0.125 + 3\times0.0625 + 4\times0.125 + 5\times0.1875 + 6\times0.125 + 7\times0.0625 = 3.25$$

4) otsu's method

For every possible threshold T, Split the image into
class C_0 : gray levels $0 \to T$
class C_1 : gray levels $T+1 \to 7$
then compute the between-class variance

$$\sigma^2(T) = w_0 w_1 (\mu_0 - \mu_1)^2$$

the optimal threshold is the one that maximize σ^2

5) compute σ^2 for each threshold

For threshold T:

$$w_0(T) = \sum_{g=0}^{T} p(g)$$

$$\mu_0(T) = \frac{\sum_{g=0}^{T} g\,p(g)}{w_0}$$

$$w_1 = 1 - w_0$$

$$\mu_1 = \frac{\mu_T - w_0 \mu_0}{w_1}$$

$$\sigma^2 = w_0 w_1 (\mu_0 - \mu_1)^2$$

Threshold T	σ^2
0	$\sigma^2 = 0.125 \times (1-0.125) \times \left(\frac{0 \times 0.125}{0.125} - \frac{3.25 - 0.125 \times 0}{1-0.125} \right)^2 = \boxed{1.51}$
1	$\sigma^2 = (0.125+0.1875) \times (1-0.3125) \times \left(\frac{1 \times 0.1875}{0.3125} - \frac{3.25 - 0.3125 \times 0.6}{1-0.3125} \right)^2 = \boxed{3.192}$
2	$\sigma^2 = 0.4375 \times 0.5625 \times (1-5)^2 = \boxed{3.938}$
3	$\sigma^2 = 0.5 \times 0.5 \times (1.25 - 5.25)^2 = \boxed{4} \leftarrow$ Max
4	$\sigma^2 = 3.504$
5	$\sigma^2 = 2.194$
6	$\sigma^2 = 0.938$

Maximum between-class Variance : $\sigma^2 = 4$
occurs at threshold : $\boxed{T=3}$
$\therefore$ optimal otsu threshold : $\boxed{T=3}$